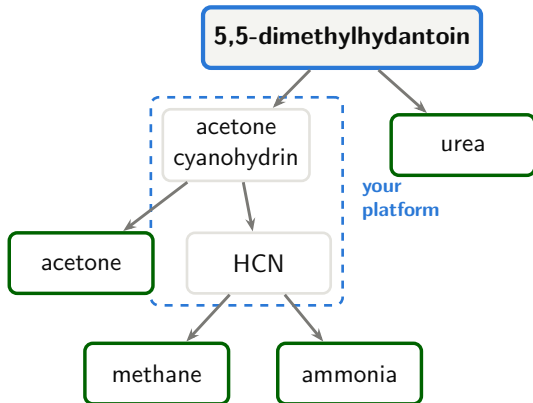


What is retrosynthesis?

Planning a synthesis *backwards* — from the molecule you want, to things you can buy



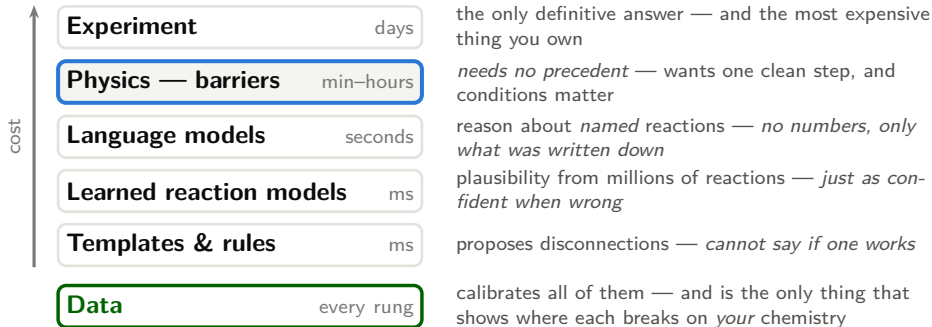
 what you want  can buy it — stop here

1. Each arrow is one reaction. Read **backwards**: *the boxes below make the box above.*
2. Stop when you reach something you can **buy**.
3. The spine is **your own value chain** — HCN to cyanohydrin to hydantoin.

The catch. Enumerating these trees is easy. Knowing which arrows *work in a reactor* is not — and that is the rest of this talk.

The tools that exist — and why none of them is enough

Each one is strong exactly where the others are blind. The opportunity is in *combining* them.



Our objective is the ladder. Send each question to the **cheapest tool that can answer it**; climb a rung only where the tools **disagree** — an observable, with no model grading its own homework.

Active learning is the research. *What* to escalate is the open question we work on — for computation, and then for **which experiment to run next**.

Where these methods pay off

Four recurring needs in industrial process chemistry — ranked by what is ready today

INDUSTRIAL NEED	WHAT THE METHODS OFFER	MATURITY
Greener or cheaper route	Route search <i>under a constraint</i> — avoid a reagent, cut cost or hazard.	High impact
Lab-to-industrial routes	Feasibility-ranked routes to a chosen target, with an explicit “cannot judge” bucket.	Most mature
Viability in real conditions	First-principles <i>barriers</i> : kinetics, not thermodynamics.	Differentiating
Feedstock valorisation	Proposing higher-value targets reachable from a cheap feedstock.	Roadmap only

The caveat we would rather raise ourselves. Public reaction models — ours included — learn from patent literature that is heavily organic and pharmaceutical, so **specialty and process chemistry is under-represented** in them. What transfers is the method, not memorised templates — which is why adapting to a producer’s own reaction records is the natural first step.